EX: 9	View and capture network traffic using Wireshark.
DATE:	

AIM

To capture, monitor, and analyze network packets in real-time using **Wireshark** in order to understand network communication and protocols.

Requirements

- Wireshark installed (Windows/Linux)
- Active internet connection
- Basic knowledge of networking protocols (TCP/IP, HTTP, DNS)

PROCEDURE:

Step 1: Install Wireshark

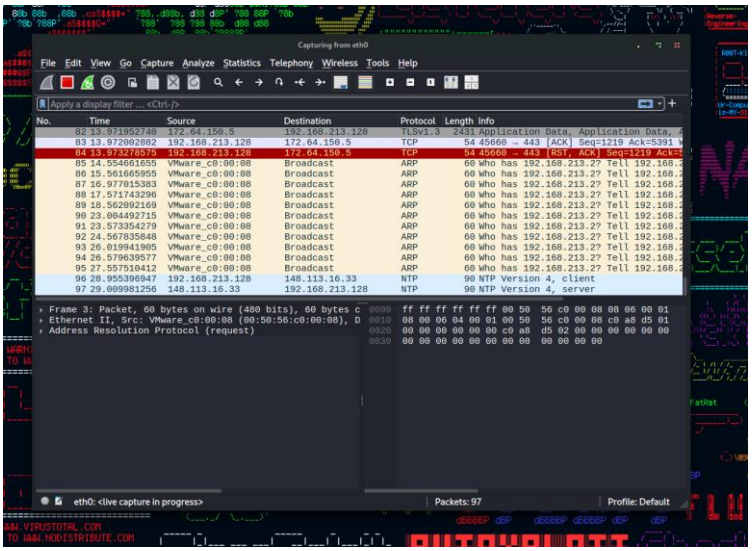
- Download Wireshark from the official website.
- Install it with default settings.
- Ensure Npcap is installed (required for packet capture).

Step 2: Launch Wireshark

- Open Wireshark.
- You will see a list of available network interfaces.

Step 3: Select Network Interface

- Choose the active interface:
  - Wi-Fi (for wireless connection)
  - Ethernet (for wired connection)



Step 4: Start Capturing

- Double-click the selected interface.
- Packet capturing will begin immediately.

Step 5: Generate Network Traffic

- Open a browser and visit websites (e.g., google.com).
- Perform actions like:
  - Search queries
  - Download files
  - Stream content

Step 6: Apply Filters

- Use display filters to analyze specific traffic:
  - http → HTTP packets
  - dns → DNS queries
  - tcp → TCP packets
  - ip.addr == 8.8.8.8 → Specific IP traffic
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Step 7: Stop Capture

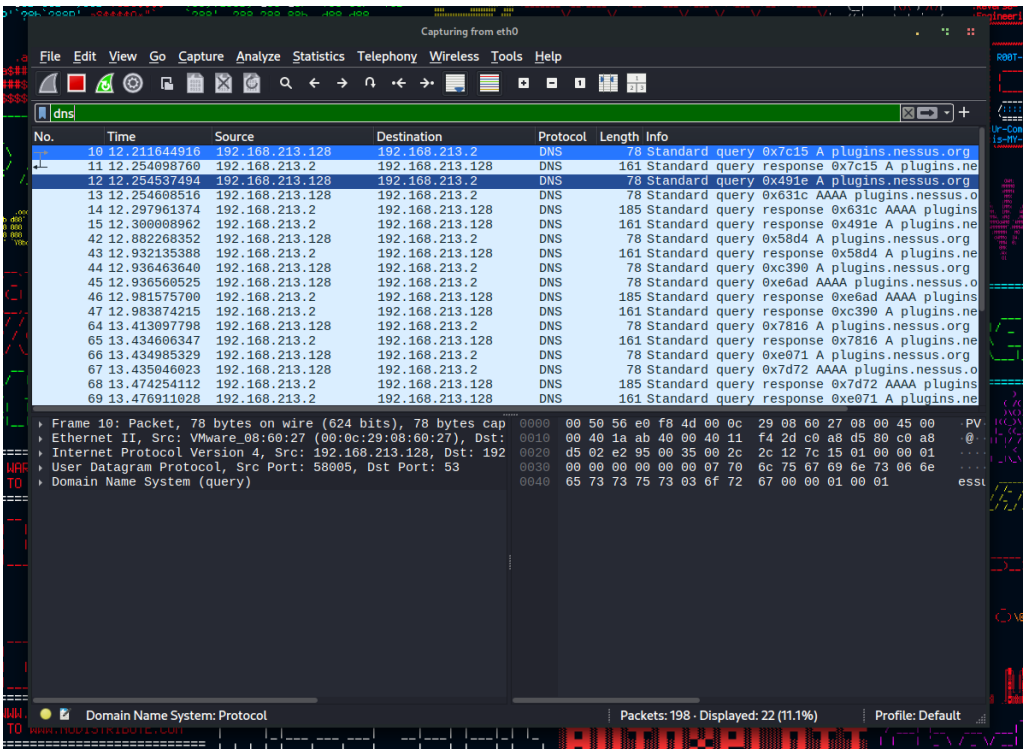
- Click the red Stop button in Wireshark.

OBSERVATION:

FILTERS USED:

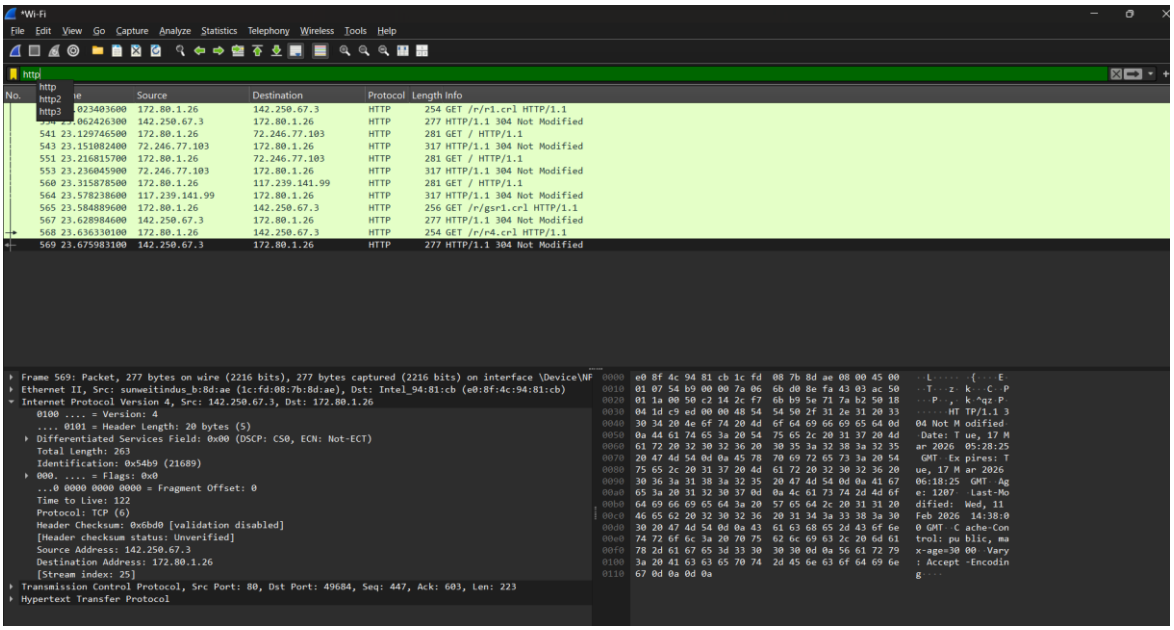
1) dns:

DNS packets were observed when the browser attempted to resolve a domain name to an IP address.



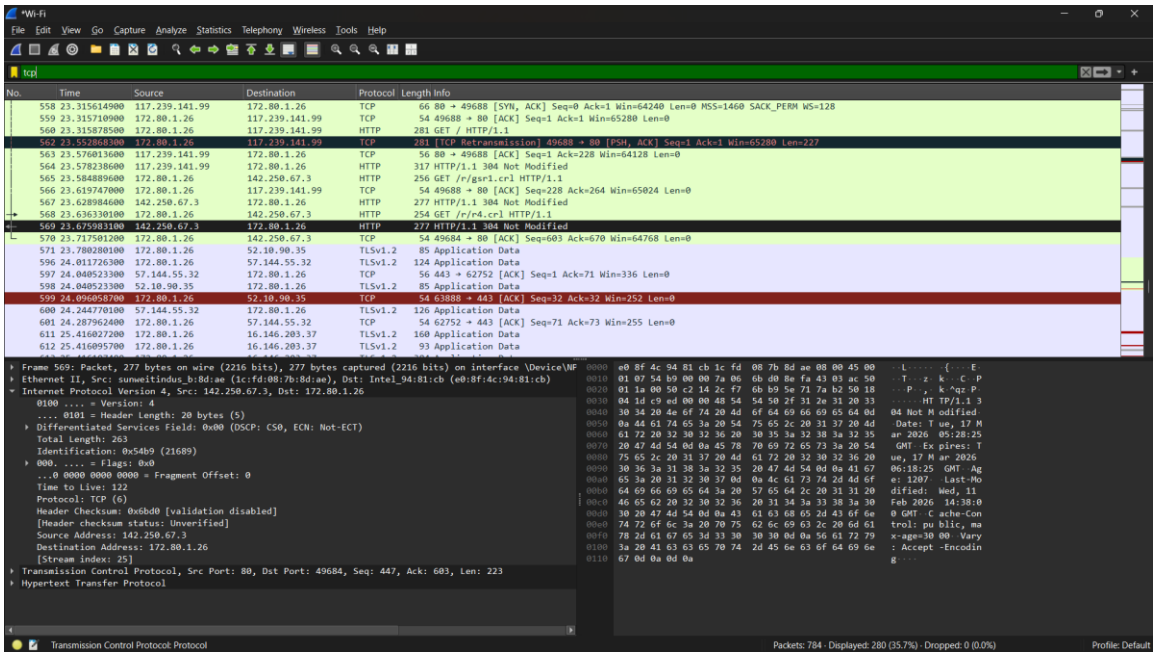
2)http:

HTTP request and response packets were captured when accessing websites.



3)tcp:

TCP packets showing connection establishment (SYN, SYN-ACK, ACK) and data transfer were observed.



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RESULT:

Network packets were successfully captured and analyzed using Wireshark. Different types of network traffic such as DNS, HTTP, TCP, and ICMP were identified using filters, demonstrating how Wireshark can be used to monitor and analyze network communication.